

Andrei Manolache

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Research interests: equivariant deep learning, training and scaling of equivariant networks, geometric models for molecules and proteins, efficient 3D perception for robots.

EDUCATION

IMPRS-IS & ELLIS; University of Stuttgart

Oct 2022 – expected early 2027

PH.D. IN ARTIFICIAL INTELLIGENCE

Ph.D. student in the IMPRS-IS and ELLIS programs, supervised by [Mathias Niepert](#) (University of Stuttgart) and co-supervised by [Karsten Borgwardt](#) (Max Planck Institute of Biochemistry). Research on inductive biases in deep learning: learned graph rewiring, training of equivariant networks, and geometric models for molecules and proteins.

University of Bucharest, Faculty of Mathematics and Computer Science

Oct 2019 – Jun 2021

M.Sc. IN ARTIFICIAL INTELLIGENCE, VALEDICTORIAN DISTINCTION

Thesis: [Deep Anomaly Detection in Text](#) (ENG), Thesis Grade 10/10

University of Bucharest, Faculty of Mathematics and Computer Science

Oct 2016 – Jun 2019

B.Sc. IN MATHEMATICS AND COMPUTER SCIENCE, FIRST CLASS HONOURS

Thesis: Autonomous Driving, Ackermann-Steered Vehicle, [Abstract](#) (ENG), Thesis Grade 10/10

RESEARCH AND WORK EXPERIENCE

MAX PLANCK INSTITUTE OF BIOCHEMISTRY | VISITING RESEARCHER

Oct 2025 – Mar 2026

- ELLIS PhD visit in Karsten Borgwardt's department in Martinsried. Built TEDBench, a large benchmark for protein fold classification, and MiAE, a self-supervised model of protein structure (ICML 2026 oral).

BITDEFENDER | RESEARCH SCIENTIST

Oct 2019 – Present

- Research scientist in the [theoretical research group](#), working on anomaly detection, natural language processing, time series and graph learning. Co-inventor of two US patents.
- Led a team of six junior machine learning engineers for six months in 2024, building internal tools such as retrieval-augmented assistants for the HR department and a transcription system for the marketing department's ticketing workflow.
- Contributed to a national and a European grant proposal. Member of the [DARWIN](#) project (€246,000, Romanian Ministry of Education) on detecting cyber-crime on the Dark Web, and involved in [ELIAS](#), the European Lighthouse of AI for Sustainability.

NXP SEMICONDUCTORS | ADAS SOFTWARE ENGINEER

May 2019 – Oct 2019

- Worked on an automotive neural network inference engine for NXP's hardware accelerator; contributed to the core C/C++ engine and explored quantization, pruning, and optimization methods.

NXP SEMICONDUCTORS | OS SOFTWARE ENGINEER INTERN

Feb 2018 – May 2019

- Built a 1/10-scale Ackermann-steered autonomous car using ROS and Linux; implemented planning, SLAM, and computer vision algorithms in Python and C++. Completed as B.Sc. thesis.

SUPERVISION AND TEACHING

- **Supervision** (senior author on each resulting paper): Boshra Ariguib, M.Sc. student at the University of Stuttgart (ICML 2026); Teodor Stupariu, high-school intern at Bitdefender (ICML 2026 workshop); Adrian C. Lutu, B.Sc. intern at Bitdefender (NeurIPS 2025 workshop oral).
- **Lecturer**, [Deep Learning course](#) that Bitdefender researchers teach at the University of Bucharest for final-year B.Sc. students. I give the opening lecture every year.
- **Teaching Assistant**: Reinforcement Learning, Foundation Models, Introduction to AI and Recent Applications of Machine Learning (Stuttgart); Deep Learning, Advanced Machine Learning and [Operating Systems](#) (Bucharest).
- **Eastern European Machine Learning Summer School**: teaching assistant in 2022 and 2024, [industry keynote speaker](#) in 2024.

PUBLICATIONS AND PATENTS

Google Scholar (Sept. 2026): 422 citations, h-index 10. ‡ junior co-author I supervised.

* equal contribution; my name is listed first, which can differ from the published order.

Conference papers

Distillation of Foundation Models for Time-dependent PDEs.

Daniel Musekamp, Boshra Ariguib, Andrei Manolache, Mathias Niepert

NeurIPS 2026

Protein Fold Classification at Scale: Benchmarking and Pretraining.

Andrei Manolache*, Dexiong Chen*, Mathias Niepert, Karsten Borgwardt

ICML 2026 Oral ($\approx 2.64\%$ of accepts)

Learning the Neighborhood: Contrast-Free Multimodal Self-Supervised Molecular Graph Pretraining.

Boshra Ariguib‡, Mathias Niepert, Andrei Manolache

ICML 2026

Learning (Approximately) Equivariant Networks via Constrained Optimization.

Andrei Manolache, Luiz F.O. Chamon, Mathias Niepert

NeurIPS 2025 Oral ($\approx 1.46\%$ of accepts)

Probabilistic Graph Rewiring via Virtual Nodes.

Andrei Manolache*, Chendi Qian*, Christopher Morris, Mathias Niepert

NeurIPS 2024

Probabilistically Rewired Message-Passing Neural Networks.

Andrei Manolache*, Chendi Qian* et al.

ICLR 2024

AD-NLP: A Benchmark for Anomaly Detection in Natural Language Processing.

Andrei Manolache*, Matei Bejan*, Marius Popescu

EMNLP 2023

VeriDark: A Large-Scale Benchmark for Authorship Verification on the Dark Web.

Andrei Manolache*, Florin Brad* et al.

NeurIPS 2022 D&B Track

AnoShift: A Distribution Shift Benchmark for Unsupervised Anomaly Detection.

Marius Dragoi*, Elena Burceanu*, Emanuela Haller*, Andrei Manolache, Florin Brad

NeurIPS 2022 D&B Track

Rethinking the Authorship Verification Experimental Setups.

Andrei Manolache*, Florin Brad* et al.

EMNLP 2022

DATE: Detecting Anomalies in Text via Self-Supervision of Transformers.

Andrei Manolache, Florin Brad, Elena Burceanu

NAACL 2021; ICML 2021 UDL Workshop

Workshop papers

How the Optimizer Shapes Learned Solutions in Equivariant Neural Networks.

Teodor Stupariu‡, Andrei Manolache

ICML 2026 Workshop on Weight-Space Symmetries

ChronoGraph: A Real-World Graph-Based Multivariate Time Series Dataset.

Adrian C. Lutu*‡, Ioana Pintilie*‡, Elena Burceanu, Andrei Manolache

NeurIPS 2025 BERT²S Workshop Oral

MolMix: A Simple Yet Effective Baseline for Multimodal Molecular Representation Learning.

Andrei Manolache, Dragos Tantar, Mathias Niepert

NeurIPS 2024 MLSB Workshop

Time Series Anomaly Detection using Diffusion-based Models.

Ioana Pintilie, Andrei Manolache, Florin Brad

ICDM 2023 AI4TS Workshop

Probabilistic Task-Adaptive Graph Rewiring.

Andrei Manolache*, Chendi Qian* et al.

ICML 2023 Workshop on Differentiable Almost Everything

Preprints

Equivariance Breaks the Learning Rate.

Andrei Manolache, Mathias Niepert

arXiv 2026

GraIP: A Benchmarking Framework For Neural Graph Inverse Problems.

Semih Cantürk, Andrei Manolache et al.

arXiv 2026

Patents

Anomaly Detection Systems and Methods.

Andrei Manolache, Florin Brad, Alexandru Novac, Elena Burceanu

US Patent 11,847,111 B2, granted 2023

Systems and Methods of Detecting Chatbots.

Florin Brad, Andrei Manolache, Elena Burceanu

US 2024/0386212 A1, pending

AWARDS AND SERVICE

- **Ph.D. funding** for three years from the International Max Planck Research School for Intelligent Systems.
- **Reviewing:** NeurIPS, ICML, ICLR, EMNLP; Best Reviewer, NeurIPS 2022–2024; Gold Reviewer, ICML 2026.
- **Best Poster Award**, Eastern European Machine Learning Summer School, 2021 and 2025.
- **1st place**, ICLR 2021 Computational Geometry & Topology Challenge ([noise-invariant topological features](#)).
- **1st place, Web/Mobile category**, PyTorch Global Summer Hackathon 2020, for the medical assistant [Q&Aid](#) ([blog post](#)); I worked on its visual question answering models.
- **1st place**, Students' Scientific Communications Session, University of Bucharest (B.Sc. 2019, M.Sc. 2021).